LOGIC DESIGN EXPERIMENT 6

實驗題目

- **LOGIC DESIGN EXPERIMENT 6: Clock Generator and Shift Register**
- Wire a 74LS14 as a clock generator.
- Connect an LED at its output, and measure its output waveform.
- Wire a 74194 as a 4-bit shift-right register, and use the output of the clock generator as the clock input of the shift register. It will be able to load 0111 first, and then shift to 1011, 1101, 1110, and back to 0111, and so on. Use LEDs to show the output of the register.

邏輯設計實驗 6：時脈產生器與移位暫存器

- 將 **74LS14** 接成一個**時脈產生器**。
- 在它的輸出端接上一顆 **LED**，並量測它的輸出波形。
- 將 **74194** 接成一個 **4-bit 右移暫存器**，並使用時脈產生器的輸出作為此移位暫存器的 clock 輸入。
- 此電路需要能夠先載入 **0111**，接著依序右移成：
  - $0111 \rightarrow 1011 \rightarrow 1101 \rightarrow 1110 \rightarrow 0111$
- 然後不斷重複循環。
- 最後使用 LED 顯示暫存器的輸出狀態。

Understanding Schmitt Triggers (Rev. A) (ti.com) SNx414 and SNx4LS14 Hex Schmitt-Trigger Inverters datasheet (Rev. C) (ti.com)

CLOCK GENERATOR

- Using 7414
- Using capacitance

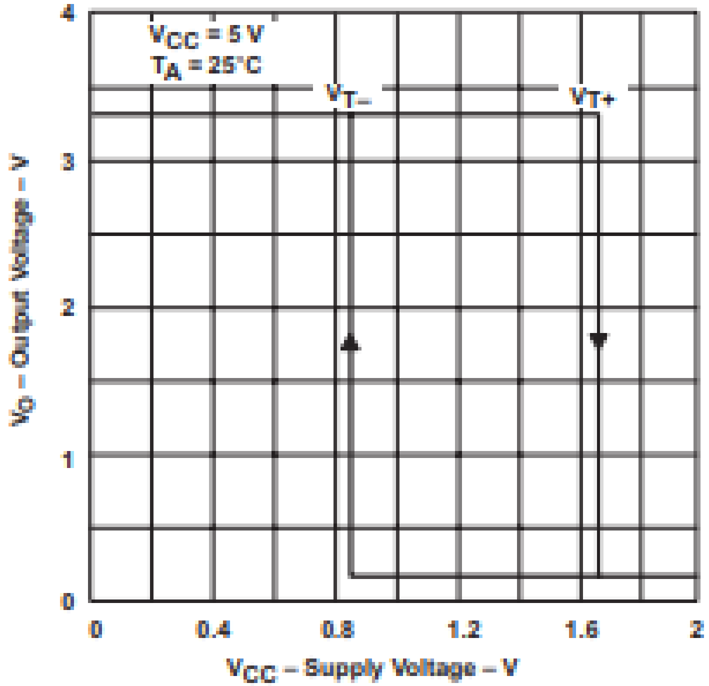


Figure 7. Output Voltage vs Input Voltage

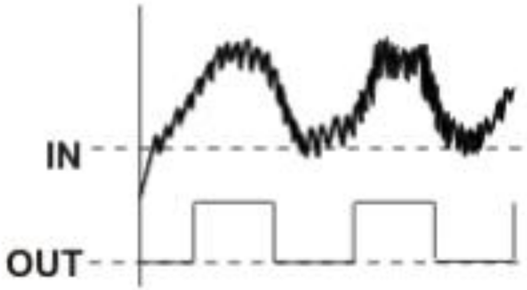


Figure 6. Clean Noisy Signals

Figure 7. Convert Slow Edges

CLOCK GENERATOR

這個 clock generator 會作為後面 **74194 移位暫存器** 的時脈輸入。
整體流程可以看成：

電阻、電容產生充放電
↓
74LS14 整形成方波
↓
輸出 clock
↓
送到 74194 控制移位

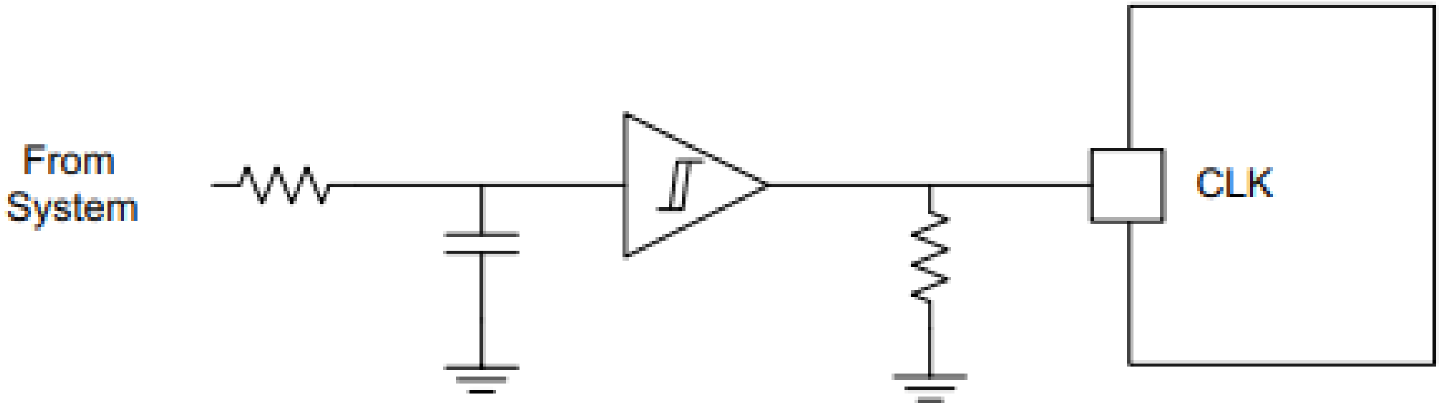
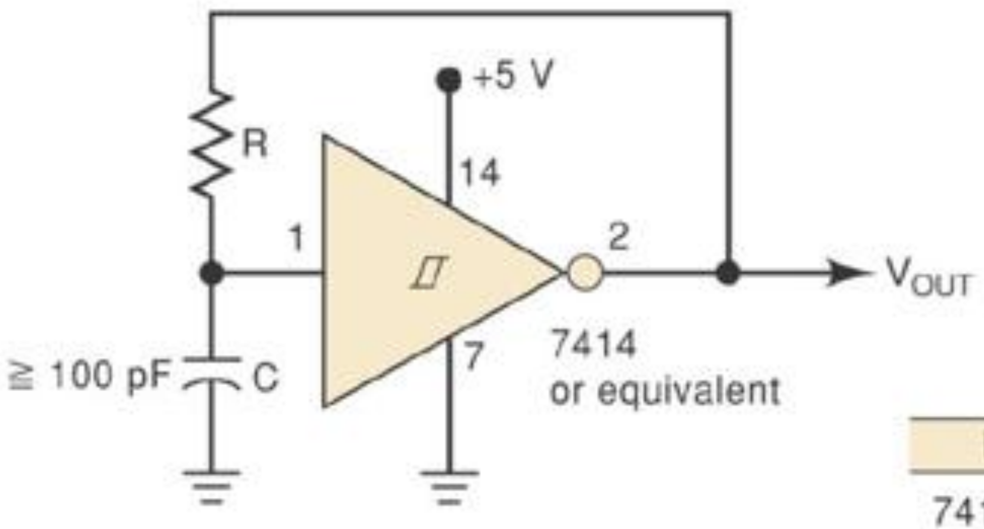
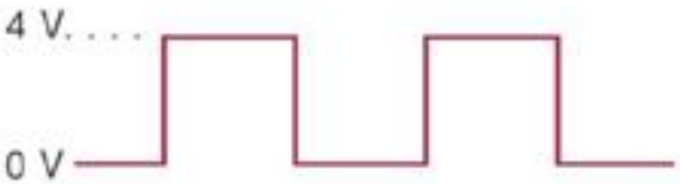


Figure 2-2. Schmitt-Trigger Clock Delay Circuit

Power-Up Behavior of Clocked Devices (Rev. B) (ti.com)

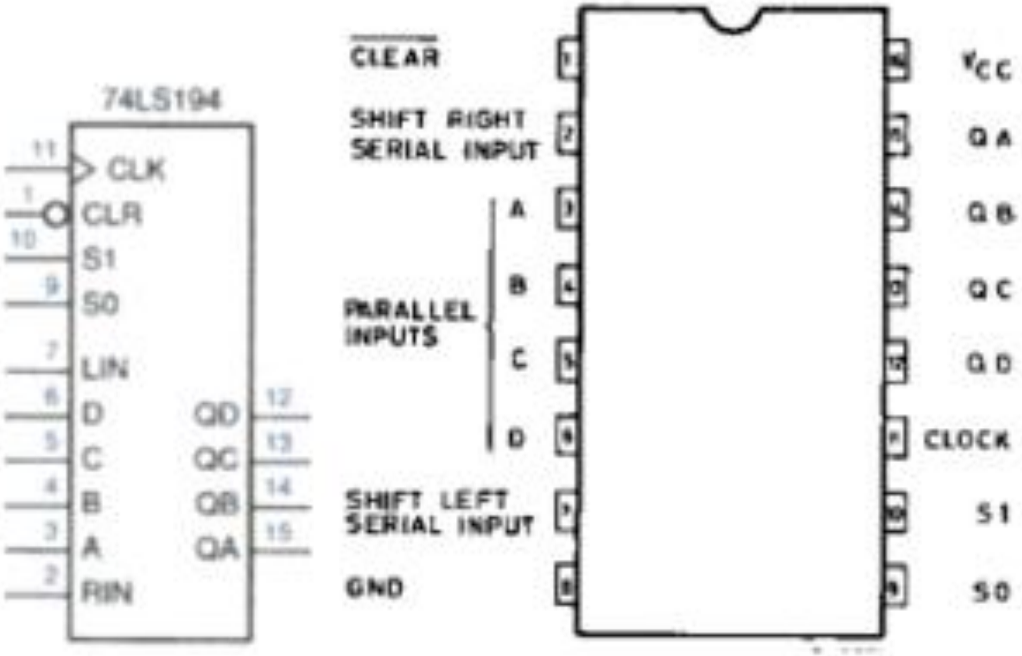
CLOCK GENERATOR





IC	Frequency	
7414	≈ 0.8/RC	(R ≦ 500 Ω)
74LS14	≈ 0.8/RC	(R ≦ 2 kΩ)
74HC14	≈ 1.2/RC	(R ≦ 10 MΩ)

74LS194


74LS194
CLK
CLR
S1
S0
LIN
D
C
B
A
RIN
QD
QC
QB
QA
CLEAR
SHIFT RIGHT SERIAL INPUT
A
B
C
D
PARALLEL INPUTS
SHIFT LEFT SERIAL INPUT
GND
VCC
QA
QB
QC
QD
CLOCK
S1
S0

74LS194

INPUTS										OUTPUS			
$\overline{\text{CLEAR}}$	MODE		CLOCK	SERIAL		PARALLEL				QA	QB	QC	QD
	S1	S0		LEFT	RIGHT	A	B	C	D				
L	X	X	X	X	X	X	X	X	X	L	L	L	L
H	X	X	↓	X	X	X	X	X	X	QA0	QB0	QC0	QD0
H	H	H	↑	X	X	a	b	c	d	a	b	c	d
H	L	H	↑	X	H	X	X	X	X	H	QAn	QBn	QCn
H	L	H	↑	X	L	X	X	X	X	L	QAn	QBn	QCn
H	H	L	↑	H	X	X	X	X	X	QBn	QCn	QDn	H
H	H	L	↑	L	X	X	X	X	X	QBn	QCn	QDn	L
H	L	L	X	X	X	X	X	X	X	QA0	QB0	QC0	QD0

感謝您

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